

AI Resume Analyzer

Intelligent Resume Screening & Job-Match Evaluation Platform

AI / ML

Python

Streamlit

Claude API

PDF Processing

OVERVIEW

Problem Statement

Recruiters often spend significant time manually reviewing resumes and comparing them against job requirements. This process can be time-consuming, inconsistent, and prone to human bias. An AI-powered solution is needed to bring speed, objectivity, and actionable insight to the hiring workflow.

GOALS

Objective

- Accept a candidate's resume in **PDF format**.
- Accept a **job description** as free-form text input.
- Extract and analyse information from both documents.
- Calculate a **Resume Match Percentage** (0-100%) showing alignment with job requirements.
- Identify missing skills, keywords, and qualifications.
- Generate **personalised improvement suggestions** using an AI model.
- Present analysis through a clean, user-friendly interface.

SPECIFICATIONS

Functional Requirements

Input	Processing
- Resume upload - PDF format - Job Description - free-form text area	- Extract raw text from uploaded PDF - Analyse resume against job description - Detect matching skills and keywords - Identify missing skills and qualifications - Evaluate relevant experience - Check education requirements

Output

Match %	0-100% alignment score between resume and job description
Matching Skills	Skills present in both the resume and job description
Missing Skills	Required skills absent from the resume
Strengths	Areas where the resume excels
Improvements	Specific gaps and areas to address
AI Suggestions	Personalised, actionable improvement recommendations
Recommendation	Strong Match Moderate Match Weak Match

ARCHITECTURE

Technical Requirements

Frontend	Backend	AI / LLM	PDF Libs
Streamlit	Python 3.9+	Anthropic Claude API OpenAI API Gemini API	PyPDF2 pdfplumber PyMuPDF (fitz)